Training a small GPT on TinyStories under a one-hour budget: model size, token exposure, and comparison limits

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Github Repository: github.com/KeremOzemre/Optimization-for-Large-Scale-Data

Abstract

We trained 20 nanoGPT models from random weights on TinyStories, each for one hour on an H100 MIG slice: five sizes (16–253M parameters) crossed with four data fractions (25/50/75/100%). XL (≈124M) has the lowest recorded validation loss among the tested families under this shared recipe. XXL processes about half as many tokens. This is a conditional fixed-time result; reduced exposure is a hypothesis for the ranking, not an established cause. Full-data exposure ranges from 0.303 to 2.077 cache-equivalents, with no guarantee of unique coverage. The original validation-selected `xl_data50` has test perplexity 4.41. A supplied common-validation audit and six fresh full-data trials (three seeds per finalist) strengthen the fixed-time observation: mean CE is 1.494657 for XL and 1.524387 for XXL, with XL lower in every trial. This is initial repeatability evidence for the shared recipe, not a significance or universal-size claim. Missing local evaluation-protocol files and checkpoints limit independent reproduction. Equal-token and capability checks remain pending; four-hour runs are deferred.

1 Compute budget declaration

Every model starts from random weights (nanoGPT initialisation, with no pretrained weights, embeddings, or tokenizer fine-tuning) and trains for approximately one hour (3600 s) of in-loop

wall-clock per model. Data preparation, the final validation pass, and checkpointing all happen outside that timed loop. All 20 runs use the same hardware: a single NVIDIA H100 NVL MIG 3g.47gb slice in `bfloat16`. The same hour of wall-clock does not buy the same number of tokens for every size, and that gap is central to interpreting our result (§5). The completed repetition study adds six one-hour training runs under the same recorded GPU allocation and training precision; offline evaluation time is additional.

2 Research question

Which model size and training-data fraction give the best language-modeling performance under a fixed one-hour budget on identical hardware? We hold the tokenizer, optimizer, schedule, context length, and effective batch size fixed, and vary only (a) model size and (b) the fraction of the training split used.

3 Experimental setup

- **Data:** `roneneldan/TinyStories`. We train on the official train split, and split the official validation set 80/20 into validation and test at the story level (seed 42). All 20 archived test results have been inspected; future model choices use validation (§5.2).
- **Tokenizer:** GPT-2 (`tiktoken`), vocabulary 50,257, with an end-of-text token after every story. Since the tokenizer never changes, per-token cross-entropy and perplexity are directly comparable across all our runs.
- **Model:** nanoGPT [3] GPT with tied input/output embeddings, causal self-attention, context length 256, and dropout 0.1.
- **Evaluation objective:** held-out cross-entropy on the test split, also reported as perplexity ($\exp$ of the loss) and bits per byte. Cross-entropy is the quantity the model is trained on, measured on unseen text, so it reads language-modeling quality directly; perplexity is its interpretable version; and bits per byte lets our numbers be compared to work that uses a different tokenizer [1]. Splitting at the story level with seed 42 keeps any single story from leaking between validation and test.

Data fractions: 25/50/75/100% of the training split, giving a 5×4 grid of 20 runs. **Optimizer and schedule:** AdamW (lr 3×10^{-4}, weight decay 0.1, betas 0.9/0.95, gradient clip 1.0), with a time-based 2% warmup and then cosine decay to 10% of the peak learning rate.

4 Methodology and reproducibility

The exact commands are in the repository README; the pipeline is five steps.

1. `run.py prepare` builds (or reuses) the token cache for a given data fraction.

Table 1: The five model sizes. Width divides evenly by the head count; context length (256) and dropout (0.1) are shared. We adjust microbatch and accumulation to keep an effective batch of 16 sequences (4096 tokens) per optimizer step at every size.

Size	Layers	Width	Heads	Params (M)	Microbatch × accum.
`small`	4	256	4	16.08	16×1
`medium`	6	384	6	30.02	16×1
`large`	8	512	8	51.04	16×1
`xl`	12	768	12	123.75	8×2
`xxl`	16	1024	16	253.09	4×4

2. Archived `run_config.json` files record the actual settings; checked-in YAMLs provide runnable configurations.

3. `run.py train` trains one config; its outputs (`run_config.json`, `metrics.csv`, `checkpoint.pt`) land in `results/`. Checkpoints and logs are mirrored to the Hugging Face Hub (`dsa4212-tinystories`).

4. The original selection used *validation* loss; all 20 test scores were subsequently exported and inspected.

5. `eval_hf.py` downloads every checkpoint and scores the full test split; `analyze.py` then aggregates all runs into `analysis/` (the tables and figures below).

Validation protocol limitation. Original validation used 16 microbatches: 65,536 targets for Small/Medium/Large, 32,768 for XL, and 16,384 for XXL. The supplied common-validation audit addresses this comparison; its offline protocol files are not yet available locally. All six repetition configs score $4 \times 64 \times 256 = 65{,}536$ in-loop targets with a fixed evaluation seed, independent of training randomness.

5 Results

Table 2 lists all 20 runs sorted by final validation loss, the metric we select on. We read three things off these runs, shown in Figures 1–3.

Best observed size under the shared recipe. Validation loss falls approximately $1.84 \to 1.64 \to 1.55 \to 1.50$ from `small` to `xl`. The best recorded XXL loss is 1.5106, compared with 1.5025 for XL. XXL has about twice as many parameters and processes roughly half as many targets. Reduced exposure, optimizer suitability, and training randomness are possible explanations; this experiment does not distinguish them. Related compute-allocation work motivates these hypotheses [2], but our five sizes and one seed do not establish a scaling law or precise optimal size.

Table 2: All 20 runs, sorted by final validation loss (the selection metric). "Tokens" is how many training tokens the run actually processed inside the one-hour budget. Test numbers are shown for completeness, but we did not use them to select the model (§5.2).

Run	Params (M)	Data%	Tokens (M)	Val loss	Val ppl	Test ppl
xl_data50	123.75	50	273.9	**1.5025**	**4.49**	4.41
xl_data100	123.75	100	279.8	1.5068	4.51	4.40
xl_data75	123.75	75	269.1	1.5099	4.53	4.43
xxl_data100	253.09	100	143.7	1.5106	4.53	4.56
xl_data25	123.75	25	283.3	1.5118	4.54	4.45
xxl_data75	253.09	75	141.1	1.5125	4.54	4.59
xxl_data50	253.09	50	148.2	1.5155	4.55	4.56
xxl_data25	253.09	25	143.2	1.5233	4.59	4.61
large_data100	51.04	100	562.4	1.5460	4.69	4.66
large_data75	51.04	75	562.0	1.5466	4.70	4.67
large_data50	51.04	50	557.9	1.5479	4.70	4.69
large_data25	51.04	25	563.2	1.5509	4.72	4.70
medium_data100	30.02	100	734.6	1.6444	5.18	5.16
medium_data75	30.02	75	738.3	1.6462	5.19	5.17
medium_data50	30.02	50	737.3	1.6527	5.22	5.20
medium_data25	30.02	25	719.1	1.6558	5.24	5.21
small_data75	16.08	75	940.8	1.8383	6.29	6.28
small_data100	16.08	100	984.3	1.8386	6.29	6.26
small_data50	16.08	50	985.5	1.8421	6.31	6.29
small_data25	16.08	25	972.0	1.8452	6.33	6.31

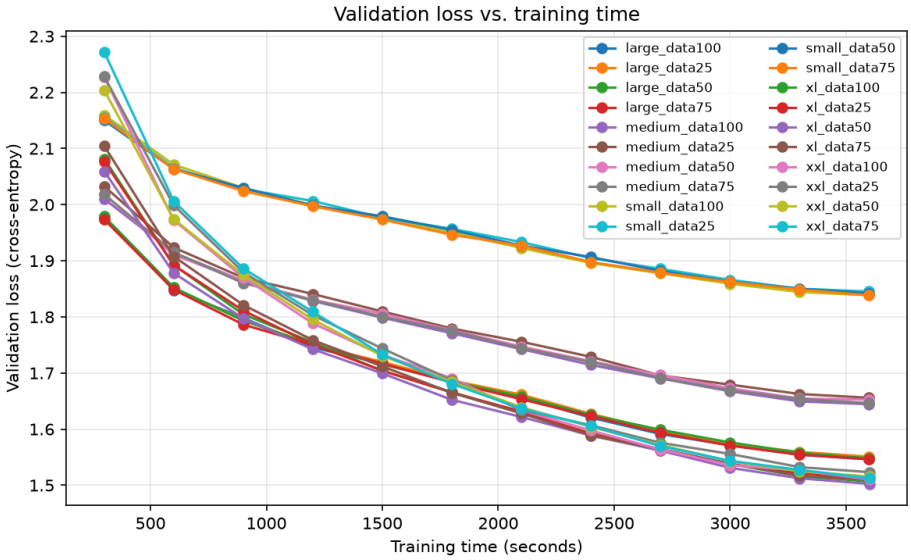


Figure 1: Original validation loss curves. Late improvements do not establish convergence. Validation sample sizes differ by model family.

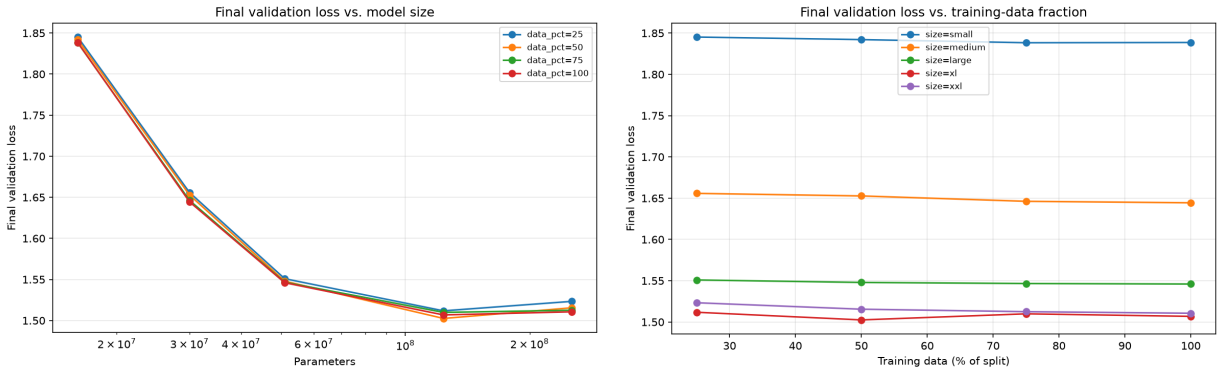


Figure 2: Recorded final validation losses by model size and data fraction. Close differences require common-target scoring and repeated training seeds.

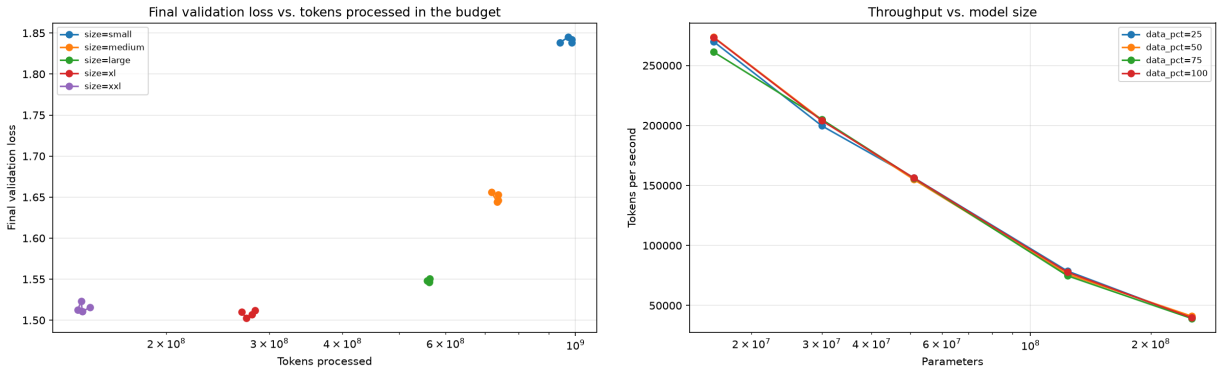


Figure 3: Processed targets and throughput under the original one-hour budget. Larger models process fewer targets; these plots do not isolate the cause of loss differences.

Small observed data-fraction effects. Within a family the observed spread is about 0.007–0.014 nats, without a consistent monotonic pattern. Nevertheless, 100% data has lower recorded validation loss than 25% in every family: a small observed benefit, not evidence that data amount is irrelevant. The 25% cache has 118,450,614 tokens and the full cache 473,992,236. Small/25% processed 8.206 cache-equivalents, contradicting a blanket explanation that training was too short to revisit data. A redundant or homogeneous corpus could contribute to the weak fraction effect, but its cause and stability across seeds are not established.

5.1 Token exposure audit

All 262,432 logged counters match completed optimizer steps times microbatch, accumulation, and context length. Table 3 shows full-data runs. Sampling is with replacement: equivalents count repeated targets and do not certify an epoch or unique coverage. Time estimates assume the original measured throughput.

Table 3: Full-cache exposure and estimated time for one equivalent.

Model	Targets (M)	Equivalents	Time for one equivalent
Small	984.326	2.077	28m 54s
Medium	734.642	1.550	38m 43s
Large	562.430	1.187	50m 34s
XL	279.831	0.590	1h 42m
XXL	143.655	0.303	3h 18m

5.2 Archived test results

The original validation selection gives `xl_data50`, with test loss 1.4846, perplexity 4.41, and 0.5237 bits per byte. Full-data XL has test loss 1.4824; full-data XXL has 1.5174. All 20 test scores

have subsequently been inspected. They must be disclosed as exploratory comparisons; future adaptive decisions use validation and cannot treat this test set as fresh confirmation. The archived full-test protocol scores 957,184 targets and drops a short tail. The new full evaluator is designed to include the tail and record that protocol change. The supplied follow-up CSVs report the intended target count, but the corresponding protocol JSON files have not been supplied locally.

We have no training repeats supporting a claim that fraction differences are "within noise" or statistically indistinguishable. Equal tokenization makes CE comparable as an objective, but identical targets and context policy are also needed for a close numerical comparison.

5.3 Qualitative observations

Existing examples show fluent TinyStories-style text even from smaller models. Some XL examples appear more coherent, but the five-prompt, single-sample, unblinded inspection is illustrative rather than a quantified capability result. Frozen prompts, two generation seeds, and blind rating sheets form a secondary follow-up. The original 20 color prompts always queried the first-mentioned object; a first-color copying rule scores 100%. Their replacement is a versioned 80-prompt diagnostic counterbalancing queried object, mention order, and color assignment. Its results are pending and would not establish general context-tracking or reasoning ability.

5.4 Exploratory size and compute plots

The proxy $C \approx 6ND$ uses parameter count N and processed targets D; it is not a hardware measurement. It does not establish GPU utilization. A quadratic fit to five size-level losses is descriptive and depends on the functional form, single seed, evaluation samples, and chosen points. The updated fit uses 100% data for every size; its vertex is not evidence of a compute-optimal parameter count.

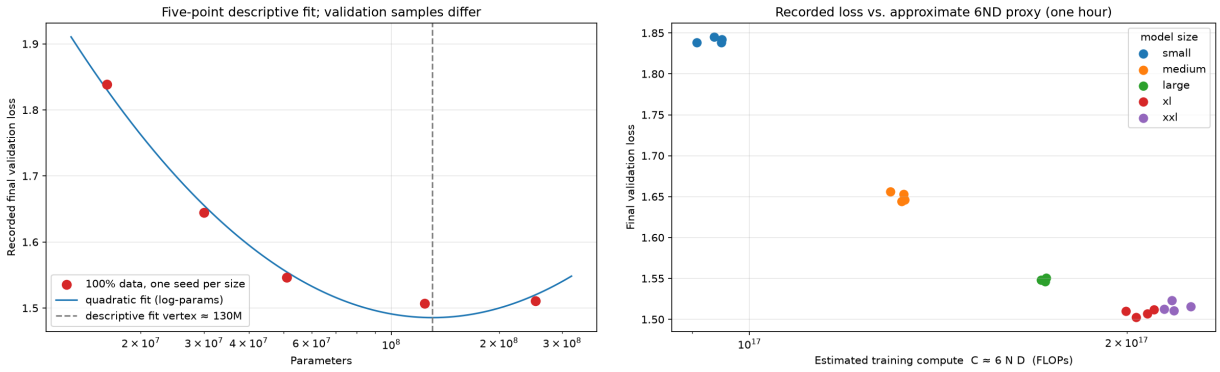


Figure 4: Descriptive size fit on full-data runs and the approximate $6ND$ proxy. Neither panel establishes an optimal size, a scaling law, or measured GPU utilization.

5.5 Additional budget and context runs: exploratory

The 30-minute Large, XL, and XXL runs have recorded validation losses 1.580, 1.633, and 1.680. Large has the lowest recorded value in that group; XL leads at one hour. However, the additional runs record `NVIDIA H100 NVL`, while the main grid records `NVIDIA H100 NVL MIG 3g.47gb`. Validation sample sizes also differ. This does not isolate a causal budget effect. The updated budget plot fixes the data fraction at 100%, but hardware still needs reconciliation.

The XL 512-token-context run records validation loss 1.4555 versus 1.5068 for the 256-token full-data baseline, and processes 190M versus 280M targets. Besides the hardware label, effective targets per optimizer step change from 4096 to 8192 and validation windows change. The observed difference cannot be attributed solely to context.

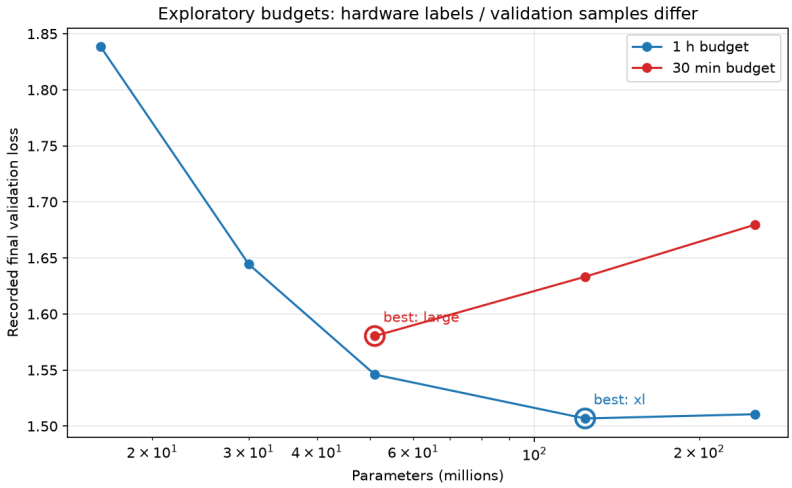


Figure 5: Exploratory recorded losses at two budgets, with 100% data. Different hardware labels and validation samples prevent a controlled budget interpretation.

5.6 Completed common-validation audit: supplied exports

The supplied `analysis/followups/archived_validation/scores.csv` reports 3,808,643 validation targets for each archived checkpoint (Table 4). The full-data XXL–XL gap is 0.034285 nats, compared with 0.003782 in the original unequal sampled validation. The audit supports XL's advantage for these checkpoints; the original small sampled gap was not the right close-comparison estimate. XL/100% also edges XL/50% by 0.001798 nats, reversing the original sampled ordering. This tiny single-seed fraction difference does not establish that full data is reliably better than 50%.

The scored count matches the intended complete validation stream except its first token. However, neither scoring directory locally contains `protocol.json`, per-story NPZ files, or `paired_comparisons.json`. The exact offline context policy, precision, validation hash, and story-level intervals cannot be independently checked from the supplied exports.

Table 4: Supplied common-validation audit of archived checkpoints.

Archived model	Validation CE	Perplexity
Large / 100%	1.546940	4.697073
XL / 50%	1.491074	4.441863
XL / 100%	1.489276	4.433885
XXL / 100%	1.523562	4.588538

5.7 Completed one-hour repetitions: three seeds per finalist

The six supplied full-data runs use the shared recipe with seeds 42, 43, and 44. Their exported scores in `analysis/followups/seed_validation/scores.csv` report 3,808,643 validation targets per checkpoint (Table 5).

Table 5: Supplied one-hour repetitions. Training targets are shown in millions; positive XXL–XL CE differences favour XL.

Seed	XL CE	XXL CE	XXL–XL	XL targets (M)	XXL targets (M)
42	1.490381	1.529798	+0.039417	271.016	138.121
43	1.495970	1.530280	+0.034309	267.911	133.325
44	1.497618	1.513084	+0.015466	267.891	146.387

XL's mean CE is 1.494657 (sample SD 0.003793), versus 1.524387 (SD 0.009792) for XXL. The mean paired difference is $+0.029731$ nats. Exponentiating mean CE gives geometric-mean perplexity 4.457805 versus 4.592329, approximately 2.93% lower for XL. Every XL run scores below every XXL run in the supplied table. Three seeds provide initial repeatability evidence, not a 95% significance claim or a universal ranking.

Saved training configurations match the intended controls and record the same hardware label, training-cache hash, code hashes, and bfloat16 precision. All 29,961 supplied log rows have consistent step-times-4096 target counters, and final losses are finite. Recorded loop durations span 3600.008–3600.101 seconds. Mean processed targets are 268.939M for XL and 139.278M for XXL, a ratio of about 1.931. Equal time still implies unequal exposure.

Between-run variation combines training randomness and throughput variability: XXL/seed44 processes about 9.8% more targets than XXL/seed43, so its lower loss cannot be attributed to the numeric seed alone. Checkpoints and `training_summary.json` are not available locally. We cannot reload the weights, reproduce offline scores, or verify separate final-evaluation/save timing here. This is analysis of supplied results, not a new local run.

6 Discussion and limitations

The original grid addresses a practical fixed-time training question. Equal time intentionally permits different token exposure; it does not establish intrinsic capacity or equal-exposure per-

formance. The supplied common-validation audit and three-seed repetitions strengthen XL's advantage under the shared one-hour recipe, but do not establish that exposure caused the gap. The next targeted explanatory comparison is the preplanned short equal-token pair with fresh models and a common step-based schedule.

The original grid has one seed per configuration; the repetition study has only three per finalist. The recipe is not tuned per architecture, training samples with replacement, throughput varies between runs, the dataset revision was not pinned, and all 20 original test scores have been inspected. Missing offline protocol/checkpoint artifacts limit independent verification. Saved training hashes and aggregate scores do not replace reproducible evaluation artifacts or establish statistical significance.

6.1 Follow-up status and next experiment

The common-validation audit and six one-hour trials are complete as supplied, with results above. Recover the missing protocol, per-story, checkpoint, and timing artifacts; the local exports do not support checking story-level intervals. Exact commands are in `FOLLOWUP_COMMANDS.md`. Remaining work is:

1. Optionally compare fresh XL/XXL models at 35,000 updates, or 143,360,000 targets, under the same step-based learning-rate schedule. Original throughputs suggest about 31 minutes for XL and 60 minutes for XXL, around 1.5 GPU-hours total. A two-hour safety cap per model marks early stops as incomplete. Equal targets do not guarantee an epoch; one cache-equivalent is not a necessary scientific milestone. The old time-scheduled XXL checkpoint is not an exact control for a new step-scheduled XL run. This pair has not been trained.

2. Use frozen-prompt blind story ratings as a modest supplement. The corrected 80-prompt color diagnostic is exploratory. These ratings and probe scores remain pending.

The four-hour XL/XXL pair remains deferred: it costs about eight GPU-hours for one seed per model and answers a separate larger-budget question. Its schedule stretches to the budget; these runs are not checkpoint continuations. A symmetric learning-rate sensitivity check or one-model experiment with smaller data subsets would likewise address separate questions.

Completed training configs fix preparation seed 42, evaluation seed 43, and 65,536 in-loop targets independently of training randomness. The observed three-seed ranking consistently favours XL, but this is still an initial repeatability check under a shared recipe; per-architecture hyperparameters were not optimized. Story-resampling intervals, once supplied, would condition on fixed checkpoints and would not replace training repeats.

7 Conclusion

Under the recorded one-hour H100 MIG budget and shared recipe, XL consistently outperforms XXL in the supplied three-seed full-data comparison: mean CE 1.494657 versus 1.524387, or about 2.93% lower geometric-mean perplexity. The common-validation archive audit supports the same

direction. This strengthens the original fixed-time observation without proving a universal best size, significance, convergence, or the cause of the gap. Offline protocol/checkpoint artifacts remain necessary for independent reproduction. Data-fraction findings still rely on the original single-seed grid. Equal-token training and capability ratings remain pending, and their outcomes are not evidence for the current conclusion.

References

[1] Ronen Eldan and Yuanzhi Li. Tinystories: How small can language models be and still speak coherent english? *arXiv preprint arXiv:2305.07759*, 2023.

[2] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

[3] Andrej Karpathy. nanogpt. https://github.com/karpathy/nanoGPT, 2023. commit 3adf61e.

A Generated story samples

The original five-prompt samples for all 20 runs are in `analysis/generations_all.md`. These examples were inspected without blinding and are illustrative, not a scored capability benchmark. New blind samples and ratings are pending.